

Entangled Photons in Quantum Optics

Achilleas Lydakis

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1 Introduction

Quantum entanglement is a foundational concept in quantum mechanics, describing the phenomenon where particles become linked such that the state of one instantly affects the other, regardless of the distance between them. In quantum optics, photons are ideal carriers of entanglement due to their low interaction with the environment and ability to travel long distances. This makes them central to emerging technologies such as quantum communication, quantum computing, and quantum cryptography[1].

This paper focuses on the generation of entangled photons specifically through spontaneous parametric down-conversion (SPDC), one of the most widely used and experimentally accessible methods for producing entangled photon pairs. In SPDC, a nonlinear crystal is used to convert a high-energy photon into two lower-energy photons that are entangled in various degrees of freedom, such as polarization or momentum.

The following sections provide an overview of the underlying physical principles of SPDC, the role of nonlinear optical materials, the geometric and polarization characteristics of photon pair emission, and the experimental configuration used to produce polarization-entangled states. Through this focused exploration, the paper aims to clarify how entanglement is realized in practice using SPDC in quantum optics.

2 Spontaneous Parametric Down-Conversion

Spontaneous Parametric Down-Conversion (SPDC) is a widely used technique for producing quantum entanglement in photon pairs. In this process, a high-energy pump photon interacts with a nonlinear optical crystal and splits into two lower-energy photons, known as the signal and idler. These photons are created simultaneously in a single quantum event, meaning that the generation of one guarantees the existence of the other. As a result, their physical properties—such as energy, momentum, and polarization—are strongly correlated due to both the conservation laws and their common origin.

When the emission paths and distinguishing characteristics of the two photons are made indistinguishable, the pair enters a quantum superposition state in which the overall system is described as a combination of multiple possible outcomes. For instance, in the case of polarization entanglement, the joint quantum state of the signal and idler photons can be written as:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|H\rangle_s |V\rangle_i + |V\rangle_s |H\rangle_i),$$

where $|H\rangle$ and $|V\rangle$ represent horizontal and vertical polarization states, and the subscripts s and i refer to the signal and idler photons, respectively. This state means that neither photon has a definite polarization on its own, but together they are in a well-defined, entangled state.

Measuring the polarization of one photon immediately determines the polarization of the other, even across large distances—a hallmark of quantum entanglement. This entanglement arises not from classical correlation, but from the nonlocal nature of quantum mechanics and the shared creation process, making SPDC a fundamental method for investigating and utilizing entangled states.

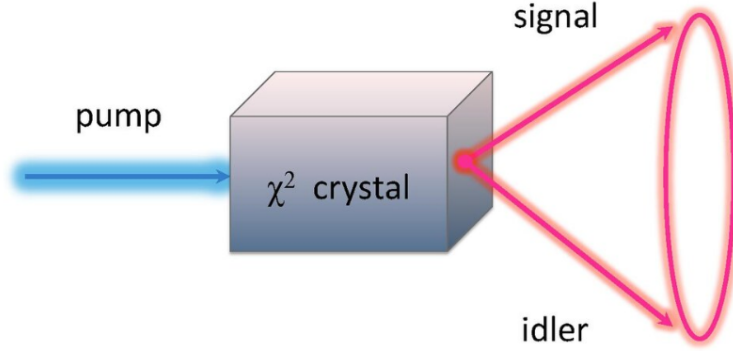


Figure 1: [4] A schematic illustration of Spontaneous Parametric Down-Conversion (SPDC), where a pump photon is converted into a pair of entangled photons within a nonlinear crystal.

Spontaneous Parametric Down-Conversion (SPDC) occurs in nonlinear optical media with a nonzero second-order susceptibility $\chi^{(2)}$, where the polarization response includes a term proportional to E^2 . The total polarization is given by:

$$\mathbf{P}(t) = \epsilon_0 [\chi^{(1)}\mathbf{E}(t) + \chi^{(2)}\mathbf{E}^2(t) + \dots].$$

In SPDC, a pump photon at frequency ω_p is spontaneously converted into two lower-energy photons—signal and idler—satisfying:

$$\omega_p = \omega_s + \omega_i, \quad \vec{k}_p = \vec{k}_s + \vec{k}_i.$$

These conservation laws ensure energy and momentum (phase matching) are preserved.

3 Emission Characteristics of Type I and Type II SPDC

There are two main configurations of SPDC: Type I and Type II.[1] In Type I SPDC, both the signal and idler photons have the same polarization, orthogonal to that of the pump. They are emitted symmetrically on the surface of a single cone around the pump beam. In contrast, Type II SPDC generates photon pairs with orthogonal polarizations. These photons are emitted on separate cones that partially overlap, producing an X-shaped or elliptical pattern in the intersection region. This difference in geometry is a direct result of the birefringent properties of the nonlinear crystal and the polarization dependence of phase matching.[2]

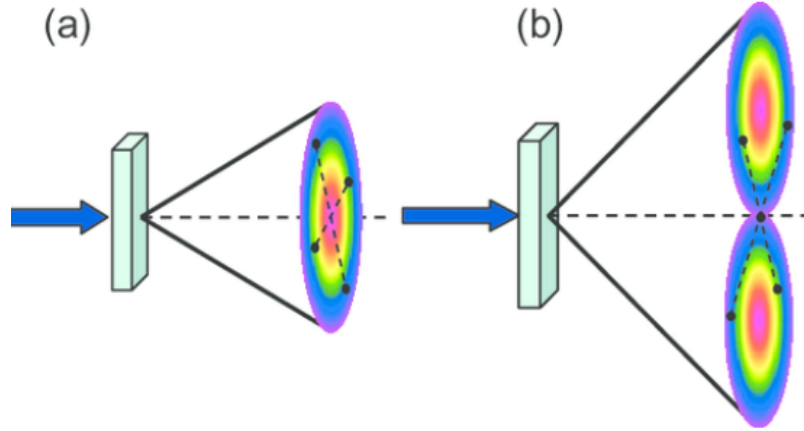


Figure 2: [5] Visualization of SPDC emission patterns. (a) Type I: signal and idler photons share the same polarization and are emitted on a single cone. (b) Type II: photons have orthogonal polarizations and emerge on separate cones with an overlapping region.

In the case of Type I SPDC, where both down-converted photons share the same polarization, the emission pattern is typically radially symmetric [2]. The photon pairs are emitted along the surface of a cone centered around the pump beam axis. This occurs because, under proper phase-matching conditions, there is a continuous range of directions that satisfy momentum conservation for the signal and idler photons. As a result, the output forms a circular ring pattern when projected onto a screen. This radial symmetry is specific to Type I processes and assumes a well-aligned, normally incident pump beam and isotropic conditions within the nonlinear crystal.

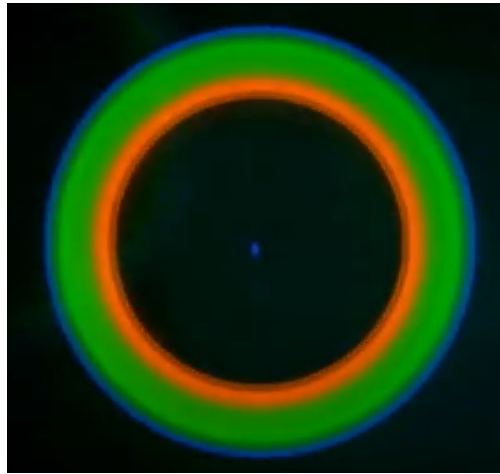


Figure 3: [2] SPDC emission pattern. The image shows how the radially symmetric emission of photon pairs in Type I SPDC forms a ring when projected onto a screen. This ring pattern results from the conical geometry of emission around the pump beam axis under phase-matching conditions.

As stated before, the conservation of momentum in Spontaneous Parametric Down-Conversion requires that

$$\vec{k}_p = \vec{k}_s + \vec{k}_i,$$

where $\vec{k}_p$, $\vec{k}_s$, and $\vec{k}_i$ are the wave vectors of the pump, signal, and idler photons, respectively. This vector relationship implies that the direction of the pump beam—specifically, the angle at which it enters the nonlinear crystal—directly influences the directions in which the signal and idler photons are emitted. In birefringent crystals, the refractive index experienced by light depends not only on its polarization but also on its propagation angle relative to the crystal’s optical axis. As a result, varying the pump beam’s angle changes the effective refractive index for the pump and alters the internal phase-matching condition. This, in turn, modifies the permissible emission angles for the signal and idler photons to satisfy both energy and momentum conservation.

Experimental observations confirm that varying the pump beam’s angle of incidence significantly affects the spatial distribution of down-converted photons. As the input angle changes, the emission pattern shifts in both symmetry and intensity, reflecting alterations in the phase-matching conditions within the crystal. These changes can lead to deformations in the SPDC emission cone, angular displacement of the output photons, and variations in the overlap region for entangled pairs. Such effects are visually observable and provide a practical demonstration of the angular sensitivity inherent to SPDC-based entangled photon sources.

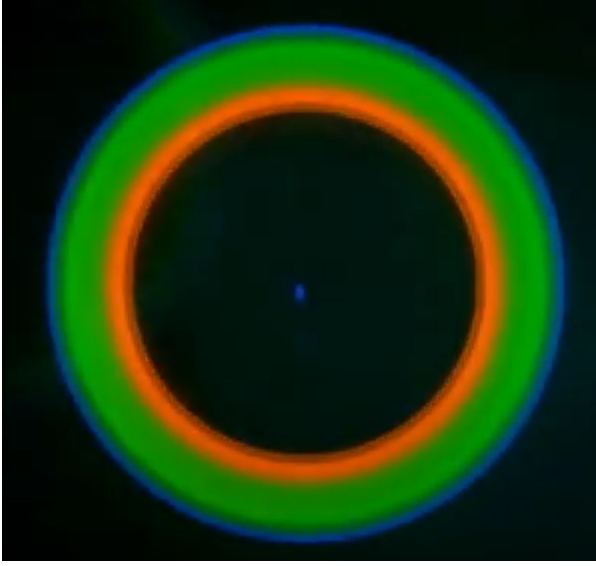


Figure 4: [2](a) SPDC emission pattern at 56° relative to the optical axis

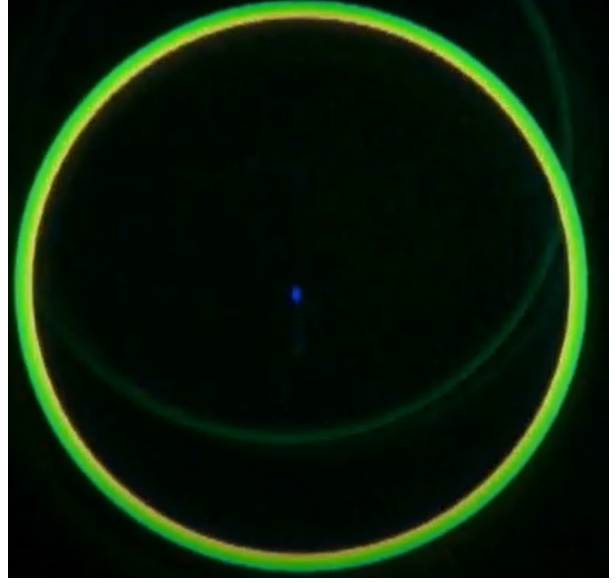


Figure 5: [2] (b) SPDC emission pattern at 59° relative to the optical axis

Figure 6: Effect of varying the pump beam angle with respect to the crystal's optical axis on SPDC emission. As the angle increases from 56° to 59°, the spatial distribution of the down-converted photons changes, illustrating the angular sensitivity of the phase-matching condition.

4 Detection in SPDC Experiments

Common $\chi^{(2)}$ materials like beta barium borate (BBO) are used due to their wide transparency range and strong nonlinearity. However, SPDC is a low-probability process—only a small fraction of pump photons that interact with the crystal result in nonlinear coupling and generate photon pairs. The majority pass through the crystal unaltered. Experimentally, the conversion efficiency is typically on the order of one successful pair generation per million to a billion pump photons [2].

And that's the reason why, the detection of the entangled photons at the output of the SPDC setup is typically carried out using photomultiplier tubes (PMTs)[2]. These serve as the primary photon detectors due to their high sensitivity. A PMT operates by using a specially designed photocathode surface that emits electrons when struck by incoming photons via the photoelectric effect. These electrons are then accelerated and multiplied through a cascade of dynodes, producing a measurable electrical pulse [6]. This amplification process enables the reliable detection of single photons, which are generated at extremely low intensities in SPDC experiments, and allows for precise time-correlation measurements essential for quantum optics.

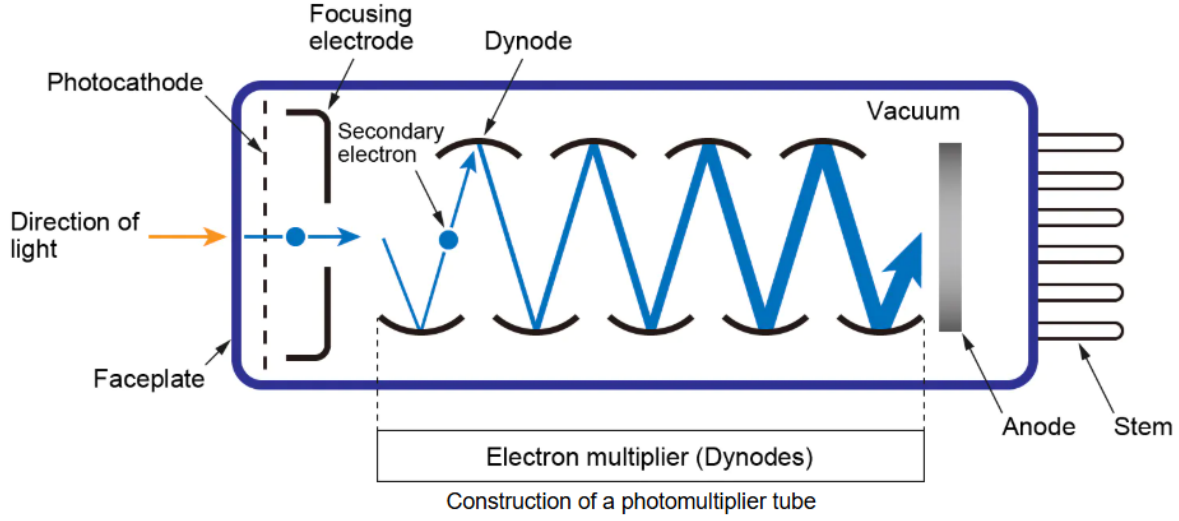


Figure 7: [6] Schematic of a photomultiplier tube (PMT). Incoming photons strike the photocathode, releasing electrons via the photoelectric effect. These are multiplied across a chain of dynodes, resulting in a detectable electrical signal.

5 Entanglement Generation

Now that we have developed a solid understanding of the components involved in SPDC, we can explore how quantum entanglement is actually achieved in practice. In typical quantum optics experiments, entanglement is encoded in the polarization degree of freedom of photons. Specifically, we use linear polarization to represent the two logical states of a quantum bit (qubit): horizontal polarization ($|H\rangle$) encodes the logical state $|1\rangle$, while vertical polarization ($|V\rangle$) represents the logical state $|0\rangle$. Diagonal polarization corresponds to a superposition of these two basis states, effectively encoding a quantum state of the form

$$|D\rangle = \frac{1}{\sqrt{2}} (|H\rangle + |V\rangle),$$

which is used to pump the SPDC crystal system and create coherent combinations of photon pair states.

The polarization state of the photons produced in SPDC depends on the orientation of the optical axis of the nonlinear ($\chi^{(2)}$) crystal. When the optical axis is aligned vertically (bottom to top), a vertically polarized pump photon can down-convert into two horizontally polarized photons, resulting in the state $|H\rangle_s |H\rangle_i$. Conversely, when the optical axis is oriented horizontally (left to

right), the same pump photon leads to a pair in the $|V\rangle_s |V\rangle_i$ state[3].

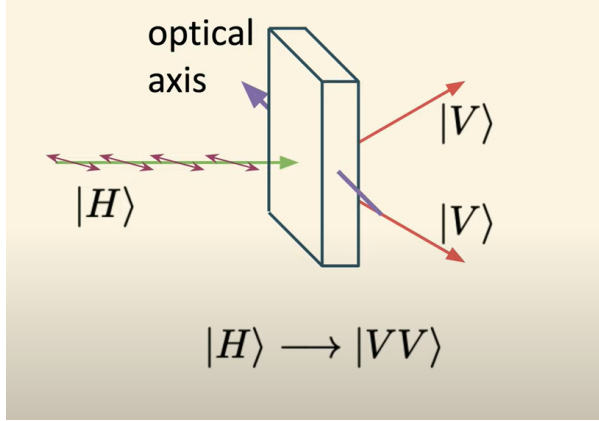


Figure 8: [3] Crystal with optical axis oriented from right to left. This orientation defines one of the emission directions used in generating polarization-entangled photon pairs.

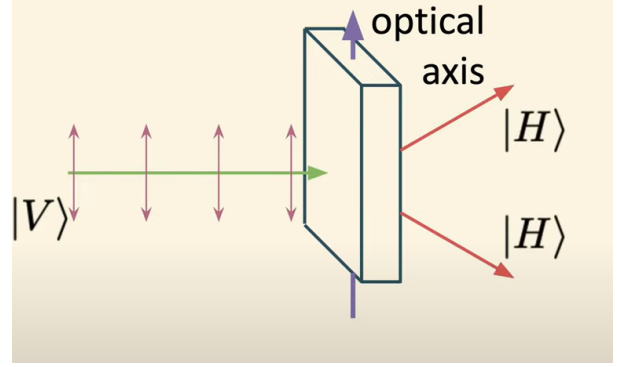


Figure 9: [3] Crystal with optical axis oriented from bottom to top. Used together with the horizontal configuration to enable coherent photon pair generation.

To create entanglement, two Type I crystals are placed consecutively, each with a different optical axis orientation—one horizontal and one vertical. When the diagonally polarized pump beam $|D\rangle$ enters this dual-crystal setup, it has an equal probability of generating down-converted photons in either crystal. If the emission paths from both crystals are made indistinguishable, the resulting quantum state is a coherent superposition of the two possibilities:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|H\rangle_s |H\rangle_i + |V\rangle_s |V\rangle_i) .$$

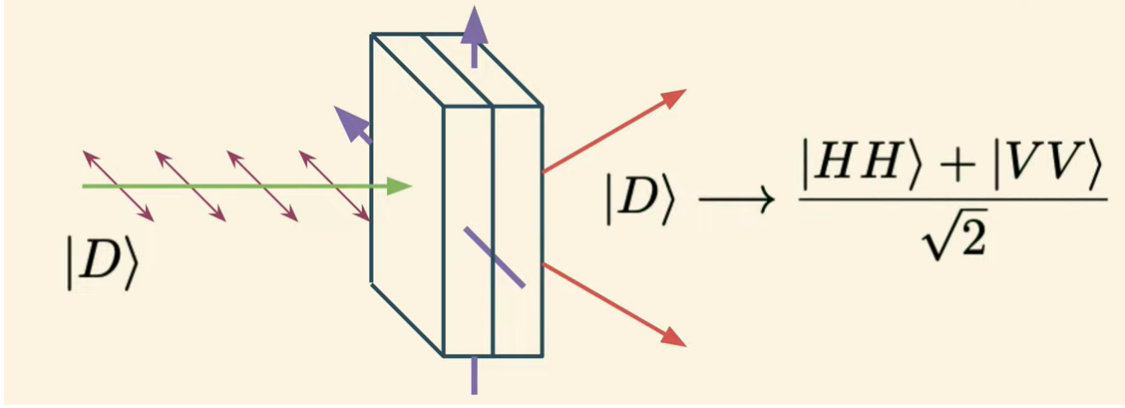


Figure 10: [3] Schematic of the full SPDC entanglement generation setup using two orthogonally oriented nonlinear crystals.

This state is one of the maximally entangled Bell states, combining both a horizontal component ($|H\rangle_s |H\rangle_i$) and a vertical component ($|V\rangle_s |V\rangle_i$). The entanglement arises from the coherence between these two emission pathways, made possible by the indistinguishability of their origin. As a result, the final light exiting the crystal setup is diagonally polarized and exhibits quantum entanglement between the two photons[3].

This understanding reinforces the earlier observation of the Type I SPDC emission pattern. The final entangled state, a coherent superposition of horizontally and vertically polarized photon pairs, naturally leads to emission along a conical surface. This cone-shaped pattern, resulting from a continuous range of phase-matched angles, is fully consistent with the physics of polarization-entangled photon generation. The diagonal polarization of the output state emerges from the indistinguishability of the emission paths, confirming the coherence and radial symmetry expected from the process.

6 Conclusion

Spontaneous Parametric Down-Conversion (SPDC) is one of the most accessible and reliable methods for generating entangled photon pairs, playing a foundational role in experimental quantum optics. By exploiting nonlinear optical processes and phase-matching conditions, SPDC enables the creation of quantum states with well-defined correlations, such as polarization-entangled Bell states. These entangled photons are essential in a wide range of quantum technologies, including quantum key distribution (QKD), quantum teleportation, quantum computing, and foundational tests of quantum mechanics such as Bell inequality violations[4]. As research continues to refine the control and efficiency of SPDC-based sources, their role in scalable quantum networks and photonic quantum information processing is expected to become even more critical.

References

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